Accurate on Average, Blind to the Student: How LLM Tutors (Fail to) Personalize Practice

Anonymous ACL submission

Abstract

To pick a practice problem a particular student will solve about 70% of the time, an LLM tutor must judge both the student and the problems. In two preregistered studies, we evaluate three open and four frontier LLMs against held-out students' real first-attempt outcomes on a primary-school mathematics platform. We identify three phenomena. First, *under-personalization*: from the student's record alone, every model's picks track ability far less than an IRT policy, so weak students get problems that are too hard and strong students problems that are too easy. Second, a *belief–action gap*: asked for probabilities, models judge students well enough to personalize, but their recommendations fall short of what those judgments imply. A simple rule using only the numbers in the prompt personalizes more strongly than every model. Third, *population anchoring*: displayed population success rates pull judgments toward the average student even when shuffled and uninformative. A simple diagnostic, $1 + \gamma s$, decomposes judgments into sensitivity to the student (γ) and item perception (s) and gives the personalization they support ($r = 0.93$ with the belief-implied policy). Average predictive accuracy and personalization can move in opposite directions, and the cost falls on the weakest students.

1 Introduction

Personalized practice is a goal of LLM tutors (Li et al., 2026): a model that reads a student's record should choose a problem at an appropriate level of challenge. This motivation is consistent with work on guided learning (Vygotsky, 1978), desirable difficulties (Bjork, 1994), and target training accuracy (Wilson et al., 2019). Whether it does is rarely measured against real students. Prior evaluations report aggregate metrics, such as how well a model predicts correctness across a dataset (Neshaei et al., 2024) or how well its difficulty ratings correlate with item statistics (Li et al., 2025). Good aggregate performance does not guarantee a personalized decision. If a tutor gives every student roughly the same problem, its predictions can still be well calibrated on average, but weak students fail too often and strong students are bored.

We study a single decision (Figure 1). Given a student's record on 100 mathematics problems and ten candidate problems the student later attempted, the model states a success probability for each candidate and recommends the one the student should solve about 70% of the time. Because the student's actual first-attempt outcome on every candidate is known, we can evaluate recommendations across ability levels, not only the accuracy of the probabilities.

Our central quantity is the *personalization slope*: the regression slope, across students, of the IRT-implied log-odds of success on the recommended problem against the student's ability θ. A perfectly personalized policy holds success at the target for everyone (slope 0); a policy that ignores the student gives everyone problems of similar difficulty, so success rises one-for-one with ability (slope 1).

Contributions.

(1) *Under-personalization.* Across two preregistered studies and seven LLMs, picks made from the student's record barely track ability, producing a success gradient that disadvantages low-ability students (§5, §6).

(2) *Belief–action gap.* Eliciting judgments and recommendations separately shows that models' own probabilities would personalize far better than their recommendations do, and that a simple rule using only the prompt's numbers personalizes more strongly than every model (§6).

(3) *Population anchoring.* Uninformative shuffled population rates lower sensitivity to the student in 6 of 7 models, so the number itself

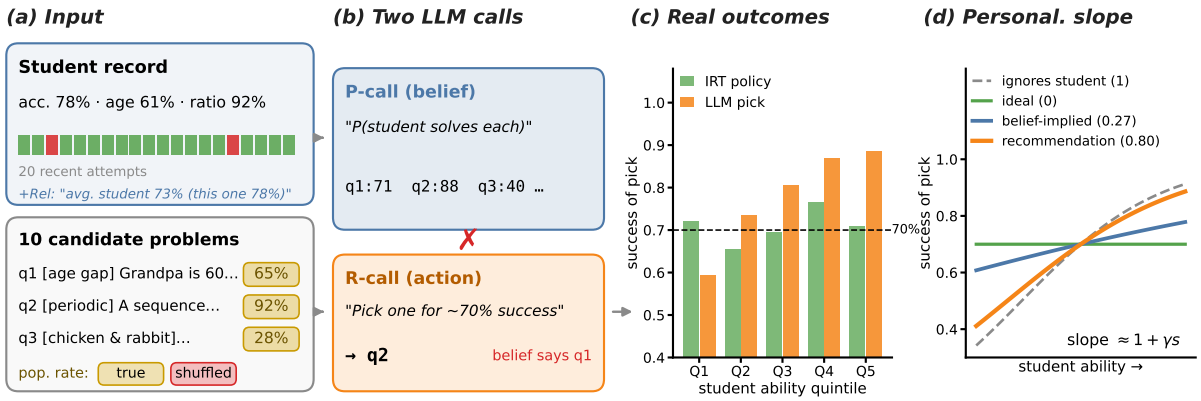


Figure 1: Overview. (a) The tutor sees a student's record and ten problems the student later attempted, optionally with population success rates (true or shuffled) and the student's relative standing. (b) It states success probabilities (P-call, its belief) and recommends one problem for about 70% success (R-call, its action); the two are elicited jointly in Study 1 and separately in Study 2. (c) Each pick is scored against the student's real first-attempt outcome; shown is Gemma-4 with the record alone versus the IRT policy (Study 1). (d) Personalization slope: 0 means success is held at the target for every student, 1 means the pick ignores the student; shown are Gemma-4's belief-implied and recommended picks with full information (Study 2).

pulls judgments toward the average student (§6).

(4) *A diagnostic.* The decomposition $1 + \gamma s$ turns stated probabilities into the personalization they support. It provides the reference for measuring the gap in (2) (§4).

2 Related Work

Student modeling and adaptive practice. Knowledge tracing predicts a student's next response from their history (Corbett and Anderson, 1994; Piech et al., 2015), and item response theory (IRT) separates student ability from item difficulty (Rasch, 1960; Lord, 1980). Target training accuracy has been studied as a way to control learning difficulty (Wilson et al., 2019). TutorLLM combines knowledge tracing and retrieval-augmented generation for personalized learning recommendations (Li et al., 2026); ExRec uses LLM-derived semantic annotations in knowledge tracing for exercise recommendation (Ozyurt et al., 2025). We use an IRT model fitted on separate students only as a reference policy and a measuring instrument, not as a competing method.

LLMs as judges of students and items. Zero-shot GPT-3.5 performed no better than naive baselines in AUC on the three datasets evaluated by Neshaei et al. (2024). Their item-difficulty estimates align poorly with students, and different models converge on a shared "machine consensus" rather than on human difficulty (Li et al., 2025). We replicate this item-level pattern (§7) without claiming it as a new finding. Our contribution concerns the *student-conditional* decision: how a model's judgments of the student and the items combine into a recommendation, and the consequences across ability levels.

Adaptive testing. Computerized adaptive testing selects items to match an examinee's current ability estimate (Wainer, 2000; van der Linden and Glas, 2010); our IRT policy is the practice analogue, and our metric measures how closely an LLM follows this classical approach.

Cognitive biases in LLMs. LLMs show human-like anchoring and framing effects in code generation (Jones and Steinhardt, 2022) and decision-making tasks (Echterhoff et al., 2024). Population anchoring is one instance: an informative number anchors the judgment and suppresses the student-specific signal. Our shuffled-rate control separates anchoring on the number from rational use of the information it carries.

LLM tutoring and verbalized probabilities. Work on LLM tutors has focused on dialogue moves and remediation quality (Macina et al., 2023; Wang et al., 2024). In question-answering tasks, verbalized probabilities can be reasonably calibrated (Tian et al., 2023), but aggregate calibration does not imply sensitivity to the conditioning variable that matters for a decision. Our anchor-

ing result is conceptually related to anchoring-and-adjustment in human judgment (Tversky and Kahneman, 1974). The distinction between average and conditional predictive quality is related to the reliability–resolution decomposition of probability scores (Murphy, 1973) and calibration across identifiable subgroups (Hébert-Johnson et al., 2018). Here, the subgroups are ability levels, and the decision requires distinguishing between candidate items for each student, rather than assessing aggregate calibration alone.

3 Task and Data

Data. XES3G5M (Liu et al., 2023) (MIT license) contains question-level response sequences of Chinese primary-school students on 7,652 mathematics problems with text, knowledge-concept labels, and response correctness. We fit a Rasch model (Rasch, 1960) $P(\text{correct}) = \sigma(\theta - b)$ on the training students only. All experiments use the separate test students.

Episodes. For each test student, the record is their first 100 interactions. The candidates are ten distinct problems the student attempted among the next 100 interactions; they are text-only, calibrated on at least 30 training students, unseen in the record, and spread evenly over difficulty. The outcome for a candidate is the student's first response to it. The student's ability $\hat{\theta}$ is a MAP estimate from the record alone, and $p_{\text{irt}} = \sigma(\hat{\theta} - b)$. For the confirmatory study, problems were excluded before any confirmatory output was seen if their training success rate was below 5% or they were split subquestions lacking their first part (Appendix A).

Advisor prompt. We make one call per student. The model sees the student's overall accuracy, accuracy on the eight most practised topics, and the 20 most recent attempts (topic, truncated text, correct/wrong), followed by the ten candidates in a per-student shuffled order. It returns an integer success probability for every candidate and recommends one problem that "should be challenging but the student should succeed about 70% of the time". We compare three prompts. RECORD is the record as described. +POP adds each candidate's population first-attempt success rate (from training students). +REL additionally instructs the model that these rates describe an average student and must be adjusted for this student, and states the student's accuracy next to the average past student's accuracy on the same 100 record problems. The development set also contains the intermediate prompt with the instruction but without the relative evidence (+ADJ; Appendix B). +POP and +REL give the model essentially the item and student information an IRT policy uses.

Models and reference policies. We evaluate seven models. Three open models run locally with llama.cpp (Q4_K_M, non-thinking mode, temperature 0.7): Qwen3.6-27B, Gemma-4-26B-A4B, and Qwen3-8B. Four frontier models run through their command-line interfaces at the interfaces' default reasoning effort and temperature (neither can be set to match the open models), with all agent tools disabled: GPT-6.1 Sol and GPT-6 Luna (Codex), and Claude Opus 5.5 and Claude Sonnet 5.5 (Claude Code). All models receive the same system prompt and user message, have no tools, and must answer in a JSON schema over the candidate ids; failed calls are retried once. Reference policies pick the candidate whose score is closest to 0.7. The scores are p_{irt} for the IRT policy and the population rate for the population policy; the random policy picks a fixed pseudo-random candidate.

Protocol. All exploration used 300 development students (22 open-model configurations including two 85% target variants, plus a rerun). Before any confirmatory call, we locked the episodes, item filter, prompts, models, analysis script (with file hashes), and four hypotheses. The confirmatory set is 1000 further students; 200 of them were run twice with the RECORD prompt. In total 9,600 confirmatory calls were made, with 1 failure. Intervals are percentile intervals from 2000 student-level bootstrap resamples, shared across cells so that contrasts are paired.

4 A Two-Factor Diagnostic of Personalization

Write the model's log-odds for student u and item i as

$$\ell_{ui} \approx \alpha + \gamma\,\theta_u + e_i, \tag{1}$$

where γ is the model's *student sensitivity* (how much its predictions move with true ability) and e_i its perceived ease of item i. A target-seeking pick selects the item whose ℓ_{ui} is closest to $\tau = \text{logit}(0.7)$, i.e. perceived ease $e^\star \approx \tau - \alpha - \gamma\theta_u$. Let s be the regression coefficient of true difficulty b on perceived ease within a student's candidate set, which we call the *item-perception slope*. It is

negative when items perceived as easier are truly easier, and its magnitude shrinks toward zero as perception becomes noisier. Its scale depends on how spread out the model's log-odds are, so $|s|$ can exceed 1 when the model compresses its numbers. γ has the inverse scale, so the product $-\gamma s$, which measures the change in chosen difficulty per unit of ability, does not depend on the model's scale. Then the chosen item's difficulty is $b \approx c + s\,e^{\star}$, so $\mathrm{d}b/\mathrm{d}\theta \approx -\gamma s$, and the realized log-odds $\theta - b$ have slope

$$\text{slope} \approx 1 + \gamma s. \tag{2}$$

For $s \leq 0$, this reduces to $1 - \gamma|s|$. For $\gamma > 0$, positive s means that items perceived as easier tend to be harder, giving a predicted slope above 1. Personalization depends on the *product* of student sensitivity and item perception. A model that perfectly ranks items but ignores the student ($\gamma = 0$) gives slope 1, as does one that tracks students but perceives items as noise ($s = 0$). Both factors are estimated from the stated probabilities and the calibrated item difficulties, without the evaluated students' outcomes. γ is the coefficient on $\hat{\theta}$ in a pooled regression of ℓ on $\hat{\theta}$ and $-b$, and s is the within-student regression of b on ℓ. Stated probabilities are clipped to $[0.02, 0.98]$ before taking logits. The observed slope is computed from the model's actual recommendations. Equation 2 is a first-order decomposition of the model's judgments: it gives the personalization those judgments support. Recommendations track it to the extent that they follow the model's own probabilities. Study 1 elicits both in one call. Study 2 (§6) uses *separate* calls, with γ and s estimated on one half of the students and applied to the other half. This measures the gap between recommendations and the judgment-implied level. In development, a naive version that ignored the shrinkage of s predicted negative slopes and failed; the shrinkage-aware form was fixed before any confirmatory data were opened.

5 Study 1: Confirmatory Results

Table 1 summarizes Study 1 for all seven models on the same 300 confirmatory students and the same joint-call prompt. The preregistered confirmatory tests use the three open models on all 1000 students; their full results are in Appendix D (Tables 6 and 5), and the numbers below come from them unless noted.

LLM tutors barely personalize. With the RECORD prompt, the three models' observed slopes are 0.79–0.93 (Table 6), close to the random policy (0.95) and the population policy (0.95), far from the IRT policy (0.09). Figure 3 shows success rates across ability levels. The IRT-implied success of the chosen problem is 0.54–0.61 for the weakest fifth of students and 0.88–0.89 for the strongest, against a target of 0.70. Observed first-attempt success shows the same pattern: it remains flat under the IRT policy (0.72 vs. 0.71), while the LLM picks produce a top-minus-bottom gap of 0.22–0.33. Averaged over models, the gap exceeds the IRT policy's by 0.29 [0.19, 0.38] (H4). The failure concerns more than predictive accuracy: AUC is 0.61–0.62 with RECORD, but the models' probabilities are also largely insensitive to the student.

The decomposition tracks jointly elicited picks (H1). Across the nine confirmatory cells, the predicted slope $1 + \gamma s$ correlates with the observed slope at $r = 0.99$ [0.95, 1.00], with mean absolute error 0.07 (Figure 2). The decomposition also tracks the ordering of the development configurations ($r = 0.91$ over the 22 open-model configurations). Deviations are systematic: when the decomposition predicts strong personalization, observed slopes are somewhat higher (e.g. 0.16 predicted vs. 0.33 observed for Qwen3.6-27B +REL). A ten-item candidate set cannot always supply an item at the desired difficulty. Simulating synthetic advisors that follow Eq. 2 exactly on the actual candidate sets gives observed slopes 0.06–0.11 above prediction in this regime, which accounts for part of the gap (Appendix F). The conclusions do not depend on using the same $\hat{\theta}$ that the IRT policy uses. Against the held-out ability θ_{ho} (correlation 0.79 with $\hat{\theta}$), all slopes shrink by measurement error, but the ordering is unchanged: RECORD cells 0.58–0.68, random picks 0.68, IRT policy 0.07.

Population anchoring (H2). Adding population rates (+POP) is what a practitioner would try first, and by average metrics it works: AUC rises by 0.15 [0.14, 0.17], to 0.77. At the same time student sensitivity *falls* by 0.12 [0.10, 0.15] on average. The drop is largest for Qwen3-8B (γ 0.30→0.06), whose probabilities become nearly the population rates. The decomposition explains why personalization barely improves even though item perception becomes far more reliable ($|s|$ roughly doubles or triples). The benefit of the larger $|s|$ is limited by the smaller γ, and success gaps under +POP are no smaller than under RECORD (Table 6). The model has the information an IRT policy uses but anchors

Model	RECORD γ	RECORD Slope	RECORD Gap	+POP − RECORD ΔAUC	+POP − RECORD $\Delta\gamma$	+REL $\Delta\gamma$	+REL Slope	Tests (Holm within model) Anch.	Rel.	Harm
Qwen3.6-27B	0.36	0.82	0.23	+0.17	-0.04	+0.31	0.39	△	◯	◯
Gemma-4-26B-A4B	0.40	0.82	0.35	+0.15	-0.14	+0.31	0.33	◯	◯	◯
Qwen3-8B	0.33	0.86	0.25	+0.16	-0.26	-0.13	0.85	◯	×	◯
GPT-6.1 Sol	0.61	0.76	0.17	+0.12	-0.00	+0.22	0.13	×	◯	△
GPT-6 Luna	0.42	0.77	0.17	+0.15	+0.04	+0.30	0.23	×	◯	△
Claude Opus 5.5	0.46	0.59	0.05	+0.12	+0.07	+0.21	0.22	×	◯	×
Claude Sonnet 5.5	0.47	0.69	0.22	+0.15	-0.01	+0.18	0.28	×	◯	◯
IRT policy		0.05	-0.05							

Table 1: Study 1 for all seven models on the first 300 confirmatory students (joint call, same prompt for all seven models). Slope: observed personalization slope; Gap: success of the pick, top minus bottom ability quintile. Tests (one-sided, Holm within model): Anch. = ΔAUC> 0 and $\Delta\gamma < 0$ for +POP; Rel. = $\Delta\gamma > 0$ for +REL; Harm = gap exceeds the IRT policy's. ◯: $p < 0.05$; △: $p < 0.10$; ×: $p \geq 0.10$.

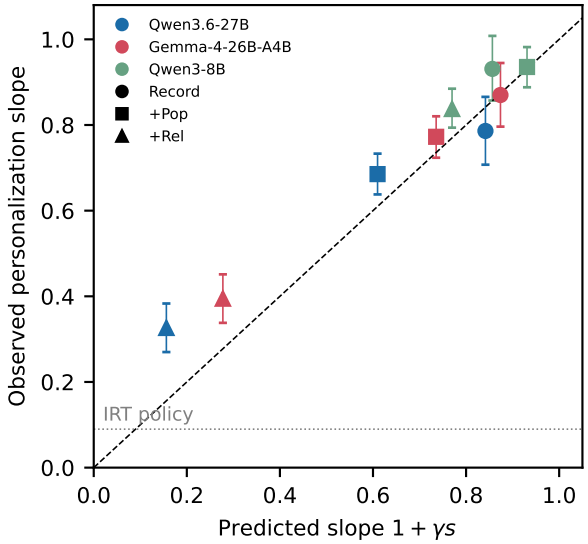


Figure 2: Study 1: predicted vs. observed personalization slope for the nine confirmatory cells (95% intervals). Dashed: identity; dotted: IRT policy. Development configurations are in Appendix B.

on the population value instead of adjusting it for the student.

Adjustment with relative evidence helps large models only (H3). Instructing the model to adjust the population rates and stating the student's accuracy next to the average student's on the same problems (+REL) raises γ relative to RECORD by 0.35 for Qwen3.6-27B and 0.29 for Gemma-4, lowering their slopes to 0.33 and 0.39. For Qwen3-8B, γ instead changes by -0.08, and the difference between the larger models and the small one is 0.40 [0.36, 0.44]. The +Rel cells of the two larger models still have success gaps of 0.15–0.18 (Figure 3). The candidate set and residual noise in both factors limit further improvement.

6 Study 2: Decoupled Calls and Controlled Information

Study 1 leaves three alternative explanations open. First, the decomposition could simply restate the probabilities used to make the pick. Second, +POP could raise $|s|$ by copying the given rates, and its effect on γ need not reflect anchoring. Third, +REL combines an instruction with information. Study 2 addresses these alternatives with a new preregistered protocol on 300 previously unused students. It was designed after Study 1's results and an external-style review were available, and it uses new prompts and new students. Its prompts, hypotheses, and analysis code were hash-locked before any of its outputs were read.

Design. For every student and prompt the model receives two independent calls. The P-call elicits probabilities only; the R-call asks only for the recommendation, which is how a tutor would be used in practice. γ and s come from P-calls of one random half of the students and predict the slope of the other half's R-calls (cross-fitting); the observed slope comes from R-calls alone. Six prompts vary the information shown while keeping the same format:

- Record.
- *Shuffled rates*: the population rates permuted across the student's candidates. The numbers are the same, but they carry no item information.
- True rates.
- True rates plus the relative-evidence sentence.

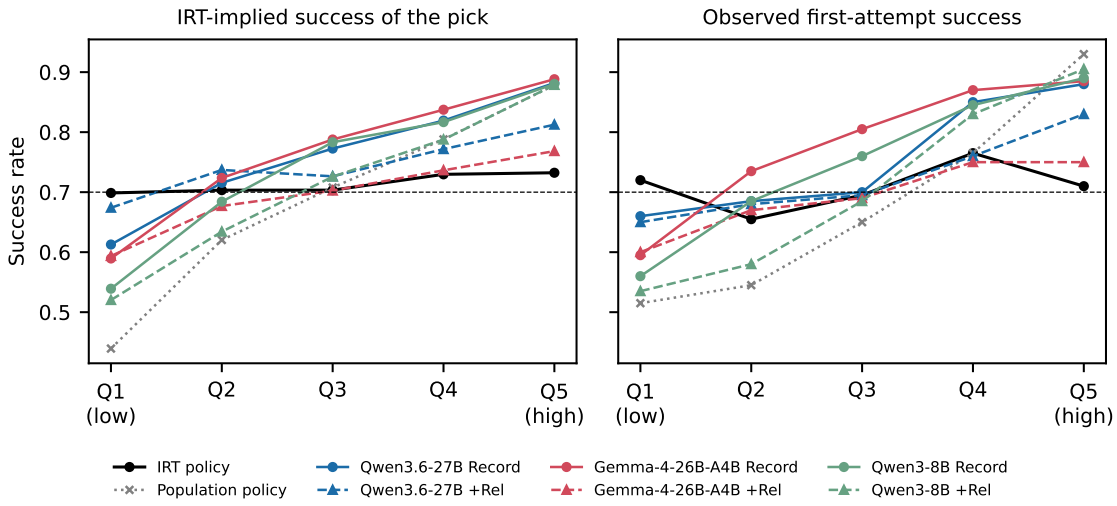


Figure 3: Success of the recommended problem by student-ability quintile (confirmatory students). Left: IRT-implied probability. Right: observed first-attempt success. Target 0.70 dashed.

- True rates, relative evidence, and the adjustment instruction.
- Shuffled rates plus the relative evidence and the instruction.

Harm is measured against a held-out ability θ_{ho}, a Rasch estimate from the student's later responses excluding the candidates, which no policy sees. Besides the IRT policy we add an *information-matched rule* that uses only numbers shown in the prompt and is not fitted. It moves each true rate by the difference between the student's and the average student's accuracy on the logit scale and picks the candidate closest to 70%. We test preregistered hypotheses with Holm correction (Holm, 1979):

- S2-H1a and S2-H1b: under-personalization relative to the IRT policy and to the information-matched rule.
- S2-H2: the multiplicative signature. Item information should lower the slope more when student evidence is present.
- S2-H3a: anchoring. Shuffled rates should lower γ.
- S2-H3b: true rates should raise AUC and lower γ.

Each model received the six prompts for 300 students (10,800 open-model and 14,400 frontier-model calls).

Table 2 gives the six tests for each of the seven models, together with the preregistered pooled test over the three open models (estimates in Appendix E).

Under-personalization and unused information (S2-H1). With the record alone, recommendation slopes exceed the IRT policy's (0.16) in all 7 models (S2-H1a, 7/7). The information-matched rule reaches a slope of 0.05 against held-out ability, close to the IRT policy (0.12). Given the same numbers, every model's recommendations are less personalized than the rule's (S2-H1b, 7/7).

Judgments versus recommendations (S2-H2). For each held-out student, the *belief-implied* policy picks the candidate whose probability in that student's P-call is closest to 70%. This measures the personalization that would result from following the model's judgments. The diagnostic predicts this level from other students' calls ($r = 0.93$; Figure 4), and the recommendations fall short of it. In the two larger open models, recommendation slopes remain high regardless of the judgments: with true rates, relative evidence, and the instruction, their belief-implied slopes are 0.24 and 0.27, while their recommendation slopes remain at 0.80–0.91. In most other models the recommendations move with the judgments across prompts ($r = 0.88$–0.98 in four of seven) but stay less personalized than the judgments allow. The loss occurs at the choice stage. Picks made in the same call as the probabilities follow them: the larger open models match their own belief-implied candidate for 0.64–0.72 of students in Study 1. In contrast, recommendation-only picks resemble noisy choices among near-tied candidates. Adding comparable choice noise to belief-implied picks produces slopes close to the observed recommendation

Model	Tests (Holm) H1a	H1b	H2	H3a	H3b	Slopes / gap Record	Belief (T+R+A)	Rec. (T+R+A)	Gap (Shuffled)
Qwen3.6-27B	◯	◯	×	◯	◯	0.93	0.24	0.91	0.43
Gemma-4-26B-A4B	◯	◯	◯	◯	◯	0.87	0.27	0.80	0.38
Qwen3-8B	◯	◯	×	◯	◯	0.85	0.76	0.82	0.32
Open models, pooled†	◯	◯	◯	◯	◯				
GPT-6.1 Sol	◯	◯	◯	◯	◯	0.73	0.17	0.20	0.57
GPT-6 Luna	◯	◯	◯	×	×	0.79	0.13	0.44	0.48
Claude Opus 5.5	◯	◯	◯	◯	◯	0.66	0.26	0.27	0.42
Claude Sonnet 5.5	◯	◯	◯	◯	◯	0.79	0.28	0.27	0.42
IRT policy						0.16		0.16	0.15

Table 2: Study 2 for all seven models (same 300 students, prompts, and analysis; six hypotheses tested within each model, Holm-adjusted). ◯: $p < 0.05$; △: $p < 0.10$; ×: $p \geq 0.10$. †The preregistered confirmatory test, pooled over the three open models (estimates in Appendix E). Slope columns are against $\hat{\theta}$; S2-H1a/b test slopes against held-out ability θ_{ho}. Record: recommendation-only slope with the record alone. Belief and Rec. (T+R+A): belief-implied and recommendation slopes with true rates, relative evidence and the adjustment instruction. Gap (Shuffled): observed success gap (top minus bottom held-out-ability quintile) with shuffled rates.

slopes. Individual picks vary between reruns, but the resulting slopes do not (Study 1: 0.82/0.73, 0.88/0.88, 0.98/0.93). Consistent with the multiplicative form, item information lowers recommendation slopes more when student evidence is present (S2-H2; 5 of 7 models and the pooled open-model test).

Anchoring on uninformative numbers (S2-H3). Shuffled rates carry no information about the candidates, yet they lower student sensitivity in 6 of 7 models (S2-H3a), and the recommendations become close to unpersonalized. True rates raise AUC while lowering γ in 6 of 7 models (S2-H3b). The belief-implied slopes show a similar pattern: shuffled rates make the judgments less personalized than the record alone in five of seven models (e.g., Claude Opus 0.39 to 0.74). Frontier models, which used true rates without losing sensitivity in Study 1, are pulled by the same numbers once they carry no information. A displayed population number thus pulls judgments toward the average student whether or not it is informative.

7 Further Observations

Item perception is a shared machine view. Averaging each model's stated probabilities per problem over the 1,843 confirmatory problems, the models agree with each other (Spearman 0.68–0.70) far more than with students' training success rates (0.18–0.20). Their perceived ease decreases with problem length ($|\rho| = 0.35$–0.44), whereas students' success rates show little association with length ($|\rho| = 0.05$). This replicates the machine-consensus finding of Li et al. (2025) and explains why $|s|$ under RECORD is low.

Recommendations are unstable. On the 200 students run twice, the same problem was recommended in 0.23 (Qwen3.6-27B), 0.35 (Gemma-4), and 0.60 (Qwen3-8B) of cases, although the stated probabilities correlated 0.73–0.86 between runs. Near-ties among candidates make the pick sensitive to sampling noise. The next problem is therefore determined partly by sampling noise, while the aggregate quantities in Eq. 2 were stable across development reruns (Appendix B).

8 Discussion

Average predictive quality is a poor proxy for the quality of personalized decisions. +POP is the clearest case: it is the prompt a practitioner would choose by AUC, yet it reduces sensitivity to the student and leaves weak students with problems that are too hard. Equation 2 gives a diagnostic that needs no outcome data. From a model's stated probabilities and item calibrations, one can estimate γ and s and predict how much personalization its judgments support. Because the two factors multiply, improving one of them yields little unless the other improves too. Study 2 adds a second, separate gap. Asked only for a recommendation, the models make far less personalized choices than their own judgments would allow. One simple mitigation is to elicit probabilities and choose by rule. In our data this belief-implied policy came close to the IRT policy once relative evidence was given. In deployment, low-ability students who most need

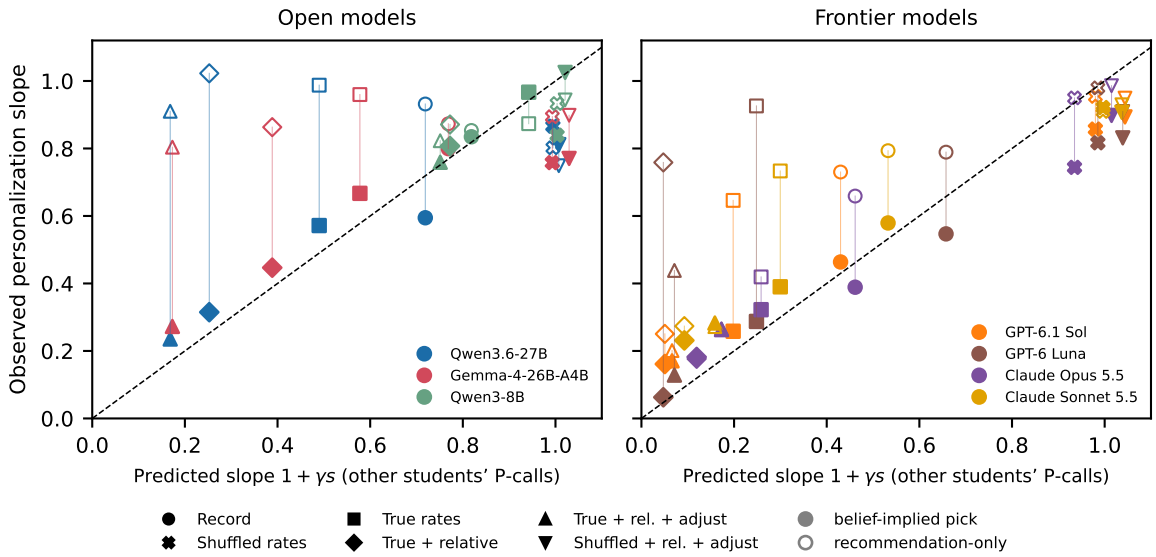


Figure 4: Study 2. Left: open models; right: frontier models. Filled markers: slope of the belief-implied pick against the diagnostic's prediction from the other students' P-calls. Hollow markers: slope of recommendation-only calls. Dashed: identity.

well-targeted practice bear most of the cost. Other mitigations are to pair LLM tutors with an explicit student model or to present relative rather than absolute evidence.

Limitations

The data come from one platform, one subject, and one country; students are primary-school learners. The outcome is first-attempt correctness, not learning gain, and the 70% target is a convention. Our development check of an 85% target is reported in the appendix but was not part of the confirmatory test. The IRT reference uses item difficulties calibrated on other students, information the RECORD prompt does not provide; +POP and +REL close most of this gap. Candidate sets are restricted to problems the student actually attempted, which keeps outcomes observed but limits the achievable personalization.

Ethics Statement

We use a public, de-identified dataset under its MIT license and collected no new human data. Our findings concern a distributional harm to low-ability students; we report them to inform the evaluation of LLM tutors, not to rank students.

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A Protocol and Item Filter

The lock file records the episode file, item filter, prompts, models, temperature, retry rule, failure rule, analysis command, the four hypotheses with their criteria, and SHA-256 hashes of the runner, analysis script, episode files, and IRT parameters. The confirmatory students are the first 1000 non-development test students ordered by a salted hash; the rerun subset is the first 200 of them. The item filter uses only item metadata and training-student rates. It removes problems with training success below 5% (often mis-keyed or multi-answer fragments) and split sub-questions that carry a marker (2)–(9) without (1), whose first part is absent. Students with fewer than ten valid probabilities are dropped from probability statistics; a missing recommendation is replaced by the random policy's candidate (one failure in total). Run order and GPU placement were changed during the confirmatory runs for speed only; prompts, seeds, and outputs are unaffected.

B Development Results

Model	Prompt	AUC	γ	$\lvert s\rvert$	Pred.	Obs.	Rerun γ / Obs.
Gemma-4-26B-A4B	Record	0.64	0.41	0.34	0.86	0.80	0.41 / 0.67
Gemma-4-26B-A4B	Record (85%)	0.62	0.41	0.31	0.87	0.72	–
Gemma-4-26B-A4B	+Pop	0.78	0.26	1.00	0.75	0.85	0.27 / 0.75
Gemma-4-26B-A4B	Recent	0.60	0.13	0.32	0.96	0.87	–
Gemma-4-26B-A4B	+Adj	0.79	0.47	1.05	0.51	0.59	–
Gemma-4-26B-A4B	+Rel	0.81	0.69	1.09	0.25	0.43	0.67 / 0.42
Gemma-4-26B-A4B	+Rel (85%)	0.80	0.72	1.05	0.24	0.34	–
Gemma-4-26B-A4B	Stats	0.62	0.47	0.25	0.89	0.82	–
Claude Opus 5.5	Record	0.69	0.45	0.77	0.65	0.58	–
Qwen3.6-27B	Record	0.63	0.34	0.48	0.83	0.76	0.36 / 0.90
Qwen3.6-27B	Record (85%)	0.63	0.41	0.47	0.81	0.74	–
Qwen3.6-27B	+Pop	0.78	0.32	1.22	0.61	0.73	0.33 / 0.63
Qwen3.6-27B	Recent	0.63	0.24	0.48	0.88	0.90	–
Qwen3.6-27B	+Adj	0.79	0.39	1.24	0.51	0.61	–
Qwen3.6-27B	+Rel	0.80	0.62	1.25	0.22	0.37	0.63 / 0.40
Qwen3.6-27B	+Rel (85%)	0.81	0.68	1.20	0.19	0.25	–
Qwen3.6-27B	Stats	0.63	0.36	0.51	0.82	0.62	–
Qwen3-8B	Record	0.63	0.29	0.53	0.85	0.67	–
Qwen3-8B	+Pop	0.78	0.07	1.10	0.92	0.85	–
Qwen3-8B	Recent	0.59	0.06	0.57	0.97	0.85	–
Qwen3-8B	+Adj	0.78	0.12	1.07	0.88	0.88	–
Qwen3-8B	+Rel	0.79	0.19	1.09	0.80	0.94	–
Qwen3-8B	Stats	0.64	0.38	0.44	0.83	0.82	–
Claude Sonnet 5.5	Record	0.66	0.44	0.64	0.71	0.63	–

Table 3: All development configurations (300 students). Recent: recent attempts only; Stats: accuracy summaries only; +Adj: +POP plus the adjustment instruction, without relative evidence. (85%): target success 85%. Rerun: second run with a different seed.

Table 3 lists every development configuration. γ and $|s|$ were stable across reruns, whereas observed slopes varied by up to about 0.15 and individual picks often changed. With an 85% target, γ and $|s|$ were essentially unchanged and the decomposition continued to track the picks, indicating that the factors describe how the model perceives students and items rather than the target.

C Quintile Table

Table 4 gives the IRT-implied and observed success of the recommended problem in each ability quintile for Study 1.

D Study 1 Confirmatory Tables

	Contrast	Estimate [95% CI]	Criterion
H1	Decomposition: r(predicted, observed slope), 9 cells	0.99 [0.95, 1.00]	>0.5 (lower)
H2a	Anchoring: ΔAUC (+Pop − Record)	0.15 [0.14, 0.17]	>0
H2b	Anchoring: $\Delta\gamma$ (+Pop − Record)	−0.12 [−0.15, −0.10]	<0
H3a	$\Delta\gamma$ (+Rel − Record), Qwen3.6-27B	0.35 [0.32, 0.39]	>0
H3b	$\Delta\gamma$ (+Rel − Record), Gemma-4	0.29 [0.24, 0.33]	>0
H3c	Large-model mean − Qwen3-8B	0.40 [0.36, 0.44]	>0
H4	Success gap (top−bottom quintile) minus IRT	0.29 [0.19, 0.38]	>0

Table 5: Study 1 preregistered tests. The criterion applies to the 95% bootstrap interval, as preregistered without multiplicity correction; all hypotheses are supported, all with intervals far from their thresholds.

Tables 6 and 5 give the preregistered Study 1 results for the three open models on all 1000 confirmatory students.

E Study 2 Details

	Contrast	Estimate [95% CI]	p_{Holm}	Result
S2-H1a	Record: slope − IRT policy slope	0.62 [0.51, 0.74]	0.002	◯
S2-H1b	True+rel.+adjust: slope − info-matched rule	0.64 [0.54, 0.75]	0.002	◯
S2-H2	Item-info effect larger with student evidence	0.08 [−0.01, 0.17]	0.037	◯
S2-H3a	γ(Shuffled) − γ(Record)	−0.24 [−0.30, −0.18]	0.002	◯
S2-H3b	ΔAUC >0 and $\Delta\gamma$ <0 (True vs Record)	0.10 [0.08, 0.13]; −0.22 [−0.28, −0.16]	0.002	◯

Table 7: Study 2 preregistered tests, pooled over the three open models (one-sided, Holm-adjusted). ◯: $p < 0.05$; △: $p < 0.10$; ×: $p \geq 0.10$.

Table 7 gives the estimates and intervals of the pooled open-model tests, and Table 8 lists every cell for all seven models.

F Held-out Ability and Candidate-set Simulation

These analyses are post hoc and descriptive. Held-out ability is a Rasch MAP estimate from each Study 1 student's responses after the record, excluding the candidates. In the simulation, synthetic advisors with log-odds $\gamma\hat{\theta} - b + \varepsilon$ ($\gamma \in [0.1, 1]$, noise sd $\in [0, 1.5]$) pick the candidate closest to 70% within each student's actual candidate set. The diagnostic predicts their slopes to within a few hundredths when personalization is weak. When the predicted slope is below 0.45, observed slopes exceed it by 0.06–0.11 because of the discrete candidate sets.

	IRT-implied success of pick, θ quintile					Observed success, θ quintile				
	Q1	Q2	Q3	Q4	Q5	Q1	Q2	Q3	Q4	Q5
IRT	0.70	0.70	0.70	0.73	0.73	0.72	0.66	0.69	0.77	0.71
POP	0.44	0.62	0.71	0.79	0.88	0.52	0.55	0.65	0.77	0.93
RAND	0.58	0.73	0.80	0.87	0.90	0.61	0.74	0.77	0.89	0.93
Qwen3.6-27B Record	0.61	0.72	0.77	0.82	0.88	0.66	0.69	0.70	0.85	0.88
Qwen3.6-27B +Pop	0.59	0.70	0.75	0.82	0.88	0.58	0.67	0.74	0.83	0.92
Qwen3.6-27B +Rel	0.67	0.74	0.73	0.77	0.81	0.65	0.68	0.69	0.76	0.83
Gemma-4-26B-A4B Record	0.59	0.72	0.79	0.84	0.89	0.59	0.73	0.81	0.87	0.89
Gemma-4-26B-A4B +Pop	0.51	0.66	0.75	0.80	0.86	0.56	0.64	0.75	0.80	0.88
Gemma-4-26B-A4B +Rel	0.59	0.68	0.70	0.74	0.77	0.60	0.67	0.69	0.75	0.75
Qwen3-8B Record	0.54	0.68	0.78	0.82	0.88	0.56	0.69	0.76	0.84	0.89
Qwen3-8B +Pop	0.46	0.61	0.71	0.78	0.87	0.53	0.55	0.69	0.79	0.86
Qwen3-8B +Rel	0.52	0.63	0.73	0.79	0.88	0.54	0.58	0.69	0.83	0.91

Table 4: IRT-implied and observed success of the recommended problem by ability quintile (confirmatory students), all cells and reference policies.

Model	Prompt	γ	$\lvert s\rvert$	$1+\gamma s$	Observed slope	AUC	Success gap
Qwen3.6-27B	Record	0.35	0.46	0.84	0.79 [0.71, 0.87]	0.62	0.22 [0.15, 0.30]
	+Pop	0.33	1.18	0.61	0.69 [0.64, 0.73]	0.77	0.34 [0.26, 0.41]
	+Rel	0.70	1.21	0.16	0.33 [0.27, 0.38]	0.80	0.18 [0.10, 0.27]
Gemma-4-26B-A4B	Record	0.39	0.32	0.87	0.87 [0.80, 0.94]	0.61	0.29 [0.21, 0.37]
	+Pop	0.27	0.97	0.74	0.77 [0.72, 0.82]	0.77	0.31 [0.23, 0.39]
	+Rel	0.68	1.07	0.28	0.39 [0.34, 0.45]	0.79	0.15 [0.06, 0.24]
Qwen3-8B	Record	0.30	0.48	0.86	0.93 [0.86, 1.01]	0.62	0.33 [0.25, 0.41]
	+Pop	0.06	1.10	0.93	0.94 [0.89, 0.98]	0.77	0.33 [0.25, 0.42]
	+Rel	0.21	1.08	0.77	0.84 [0.79, 0.88]	0.79	0.37 [0.29, 0.45]
IRT policy					0.09		-0.01 [-0.10, 0.08]

Table 6: Confirmatory cells (1000 students). γ: student sensitivity; $|s|$: magnitude of the item-perception slope; $1+\gamma s$: predicted personalization slope using signed s; observed slope of the recommended item's IRT log-odds on ability (0 = fully personalized, 1 = not personalized; random picks give 0.95). Success gap: observed first-attempt success of the recommended item in the top minus bottom ability quintile. Brackets: 95% bootstrap intervals.

G Prompt

System: "You are an experienced primary-school math tutor deciding what one specific student should practice next. Base your judgement only on the student's record shown. Output JSON only." The user message contains the record (overall accuracy, accuracy by topic, recent attempts), the ten candidates, and the two requests quoted in §3. +POP appends "(past students answered correctly on first attempt: X%)" to each candidate. +REL further adds the instruction "The past-student rates describe an average student; adjust each probability up or down for this particular student based on the record." and the sentence "On these same 100 questions, the average past student answered X% correctly on first attempt (this student: Y%)." Full prompts for every call are stored with the outputs.

Model	Prompt	γ	$\lvert s\rvert$	AUC	Pred.	Obs.	Obs. (held-out θ)	Gap	Belief–action
Qwen3.6-27B	Record	0.60	0.47	0.65	0.72	0.93	0.76	0.37	0.11
	Shuffled rates	0.43	0.01	0.59	0.99	0.80	0.65	0.43	0.17
	True rates	0.43	1.17	0.76	0.49	0.99	0.83	0.37	0.17
	True + relative	0.60	1.23	0.77	0.25	1.02	0.86	0.57	0.17
	True + rel. + adjust	0.69	1.20	0.77	0.17	0.91	0.77	0.42	0.13
	Shuffled + rel. + adjust	0.70	0.01	0.63	1.01	0.75	0.63	0.37	0.19
Gemma-4-26B-A4B	Record	0.63	0.37	0.64	0.77	0.87	0.72	0.43	0.11
	Shuffled rates	0.41	0.02	0.58	0.99	0.89	0.79	0.38	0.18
	True rates	0.44	0.96	0.74	0.58	0.96	0.76	0.43	0.25
	True + relative	0.63	0.98	0.76	0.39	0.86	0.70	0.37	0.16
	True + rel. + adjust	0.75	1.10	0.77	0.17	0.80	0.65	0.40	0.22
	Shuffled + rel. + adjust	0.78	0.04	0.62	1.03	0.90	0.77	0.42	0.14
Qwen3-8B	Record	0.35	0.52	0.63	0.82	0.85	0.76	0.42	0.11
	Shuffled rates	0.02	0.12	0.51	1.00	0.93	0.80	0.32	0.23
	True rates	0.05	1.10	0.73	0.94	0.87	0.74	0.32	0.27
	True + relative	0.21	1.11	0.75	0.77	0.87	0.72	0.37	0.24
	True + rel. + adjust	0.23	1.09	0.75	0.75	0.82	0.66	0.37	0.20
	Shuffled + rel. + adjust	0.22	0.10	0.55	1.02	0.94	0.79	0.35	0.18
GPT-6.1 Sol	Record	0.81	0.70	0.70	0.43	0.73	0.61	0.37	0.15
	Shuffled rates	0.71	0.03	0.64	0.98	0.95	0.81	0.57	0.27
	True rates	0.71	1.13	0.79	0.20	0.65	0.49	0.22	0.32
	True + relative	0.87	1.09	0.79	0.05	0.25	0.19	0.23	0.50
	True + rel. + adjust	0.87	1.08	0.79	0.07	0.20	0.15	0.18	0.55
	Shuffled + rel. + adjust	0.84	0.05	0.64	1.04	0.95	0.80	0.37	0.61
GPT-6 Luna	Record	0.61	0.57	0.68	0.66	0.79	0.67	0.40	0.16
	Shuffled rates	0.65	0.02	0.65	0.99	0.98	0.80	0.48	0.23
	True rates	0.60	1.24	0.77	0.25	0.93	0.77	0.35	0.25
	True + relative	0.81	1.18	0.78	0.05	0.76	0.59	0.38	0.27
	True + rel. + adjust	0.83	1.12	0.79	0.07	0.44	0.32	0.25	0.42
	Shuffled + rel. + adjust	0.80	0.05	0.64	1.04	0.91	0.74	0.35	0.42
Claude Opus 5.5	Record	0.58	0.93	0.72	0.46	0.66	0.60	0.28	0.21
	Shuffled rates	0.53	0.12	0.65	0.93	0.95	0.83	0.42	0.43
	True rates	0.55	1.34	0.79	0.26	0.42	0.31	0.13	0.46
	True + relative	0.69	1.27	0.80	0.12	0.18	0.13	0.17	0.59
	True + rel. + adjust	0.67	1.24	0.80	0.17	0.27	0.20	0.13	0.62
	Shuffled + rel. + adjust	0.65	0.02	0.64	1.01	0.99	0.81	0.32	0.59
Claude Sonnet 5.5	Record	0.64	0.73	0.70	0.53	0.79	0.67	0.30	0.19
	Shuffled rates	0.52	0.01	0.62	1.00	0.91	0.82	0.42	0.33
	True rates	0.55	1.28	0.79	0.30	0.73	0.59	0.37	0.32
	True + relative	0.73	1.25	0.79	0.09	0.27	0.20	0.10	0.54
	True + rel. + adjust	0.70	1.21	0.79	0.16	0.27	0.20	0.17	0.56
	Shuffled + rel. + adjust	0.65	0.06	0.62	1.04	0.93	0.74	0.35	0.55
IRT policy						0.16	0.12	0.15	
INFO policy						0.10	0.05	0.18	
POP policy						0.96	0.79	0.40	
RAND policy						0.86	0.74	0.42	

Table 8: Study 2 (300 fresh students; separate P- and R-calls). Pred.: $1 + \gamma s$, with γ and signed s from P-calls of the other student half. The $|s|$ column reports magnitude only. Obs.: slope of the R-call pick's IRT log-odds on $\hat{\theta}$; the held-out version uses θ_{ho}. Gap: observed success, top minus bottom held-out-ability quintile. Belief–action: share of R-call picks equal to the candidate closest to 70% by the same student's P-call. Frontier rows are computed with the same estimators; policies use the same students. INFO: information-matched rule. Gap uses held-out-ability quintiles.